

DERBY FIELD, NV, USA

WMO: 725805

Lat: 40.068N

Lon: 118.569W

Elev: 3902

StdP: 12.74

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24172

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.1	11.6	-2.5	5.6	21.7	1.6	6.9	24.7	26.2	48.0	21.6	43.7	2.6	110	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.5	100.5	61.8	98.1	61.1	95.5	60.2	66.2	90.4	63.7	90.0	62.0	89.3	8.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	81.7	76.1	54.0	71.6	74.4	50.6	63.1	71.6	32.9	89.5	30.7	89.9	29.5	89.1	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.3	17.5	-1.0	104.9	6.5	3.0	-5.7	107.1	-9.6	108.8	-13.2	110.5	-18.0	112.7
			-2.0	69.7	6.3	6.7	-6.6	74.5	-10.3	78.4	-13.9	82.2	-18.5	87.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	33.0	37.5	44.3	50.2	59.6	69.2	77.4	74.1	64.7	51.6	39.7	31.3	(6)
	DBStd	16.96	7.49	6.85	6.87	7.32	7.77	7.16	4.85	4.82	6.92	7.23	7.62	8.17	(7)
	HDD50	2143	529	350	199	88	12	0	0	0	3	66	316	580	(8)
	HDD65	5407	994	769	642	445	203	42	0	3	87	418	760	1044	(9)
	CDD50	3165	1	1	22	94	310	575	848	747	445	116	6	0	(10)
	CDD65	952	0	0	0	1	36	166	383	285	79	2	0	0	(11)
	CDH74	18852	0	0	11	142	986	3247	6773	5312	2151	230	0	0	(12)
	CDH80	10338	0	0	0	28	361	1671	4150	3070	1005	53	0	0	(13)
(14) Wind	WSAvg	6.4	4.6	6.0	7.2	8.1	8.1	8.2	7.3	6.5	5.9	5.5	4.9	4.6	(14)
(15) Precipitation	PrecAvg	5.3	0.6	0.5	0.4	0.5	0.6	0.5	0.2	0.3	0.4	0.3	0.4	0.5	(15)
	PrecMax	12.0	1.8	2.4	1.7	2.2	2.9	2.0	1.5	1.8	1.5	1.7	1.7	1.2	(16)
	PrecMin	3.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	1.9	0.5	0.5	0.4	0.4	0.7	0.6	0.3	0.4	0.4	0.4	0.4	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	65.0	75.2	83.8	93.1	100.2	105.3	102.1	96.3	85.7	70.2	59.1	(19)
		MCWB	46.1	46.6	49.5	53.7	59.1	60.8	63.6	61.7	59.5	56.2	49.7	46.0	(20)
	2%	DB	53.6	60.3	70.7	78.1	88.3	96.2	102.1	99.2	92.9	80.4	65.5	54.1	(21)
		MCWB	42.8	45.2	48.1	51.5	56.4	60.2	62.6	60.9	58.3	53.9	47.9	43.8	(22)
	5%	DB	49.9	56.5	66.3	73.6	84.0	93.0	99.4	96.8	89.9	75.5	61.1	49.8	(23)
		MCWB	41.0	43.5	46.4	49.6	55.1	58.9	61.7	60.1	57.6	51.6	45.6	41.1	(24)
	10%	DB	46.3	52.3	62.2	69.3	79.9	89.9	96.9	94.2	86.4	71.4	56.1	45.6	(25)
		MCWB	39.0	41.6	44.6	48.3	53.4	57.7	61.3	59.2	56.7	50.0	43.7	38.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	49.4	51.9	55.9	61.0	65.6	71.6	70.7	64.6	58.2	51.5	48.1	(27)
		MCDB	55.2	60.6	69.6	78.2	87.0	92.3	98.1	91.1	83.5	79.0	66.5	57.2	(28)
	2%	WB	44.4	46.8	49.4	52.7	58.0	62.2	67.2	65.0	61.9	55.2	49.0	44.7	(29)
		MCDB	51.2	57.8	66.9	74.4	83.2	90.6	94.7	86.8	84.3	76.2	62.2	52.4	(30)
	5%	WB	41.9	44.4	47.4	50.7	56.1	60.1	64.4	62.5	59.6	53.1	46.8	41.8	(31)
		MCDB	48.3	54.7	64.0	71.0	80.3	88.2	91.0	88.9	83.7	71.8	59.2	48.5	(32)
	10%	WB	39.5	42.2	45.5	48.7	54.4	58.6	62.6	60.9	57.6	51.0	44.4	38.8	(33)
		MCDB	45.8	51.5	60.3	67.1	76.6	86.3	90.7	88.5	83.3	68.5	55.4	44.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	23.7	26.0	30.4	29.9	30.5	33.9	37.5	38.5	38.1	34.6	29.4	23.9	(35)
		MCDBR	29.6	32.3	39.0	39.6	38.4	39.6	40.6	42.0	43.6	42.4	36.9	27.7	(36)
	5% WB	MCWBR	20.7	19.9	21.3	19.7	16.0	16.0	15.4	16.3	19.1	22.0	22.2	19.2	(37)
		MCWBR	25.0	28.8	35.1	35.6	34.9	37.0	36.7	37.0	38.1	37.3	33.3	25.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.252	0.259	0.270	0.293	0.317	0.310	0.321	0.312	0.293	0.269	0.254	0.250	(40)
		taud	2.643	2.614	2.579	2.487	2.439	2.486	2.476	2.488	2.536	2.625	2.633	2.635	(41)
	Ebn at Noon		300	311	316	312	304	305	301	302	303	303	296	292	(42)
		Edh at Noon	21	24	28	33	36	34	34	33	29	24	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		712	1039	1472	1918	2252	2587	2525	2257	1853	1301	845	598	(44)
	RadStd		108	94	101	117	179	123	107	99	56	85	47	80	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	+158	+83

Nomenclature: See separate page